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G H Raisoni International Skill Tech University, Pune

S.Y. B. Tech (AI/AIDS/DS/AIML)

TERM-III (Regular)

CAE II- WINTER 2025 (2024 Pattern)

DATA STRUCTURES AND ALGORITHMS (25BTPCL302)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Differentiate between stack and queue	[5]	3	2
b)	Solve Tower of Hanoi for Disks =3	[5]	3	3
	OR			
Q 2.(a)	Define circular queue. Why is it needed?	[5]	3	1
(b)	Convert Infix to Postfix (A+B/C−(D∗E)+G)	[5]	3	5
Q 3.(a)	Differentiate between Skewed Tree and Sub tree	[5]	4	2
(b)	Draw a tree with root A, children B and C, and B having two children D and E. Identify the root, leaf nodes, and degree of each node	[5]	4	4
	OR			
Q. 4 (a)	For the following tree, state its height, degree, and number of leaf nodes **A** **/ ** **B C** **/ \ ** **D E F**	[5]	4	2
(b)	Create a tree with at least 3 levels and identify: ○ Root node ○ Leaf nodes ○ Non-leaf nodes ○ Degree of each node ○ Height of the tree	[5]	4	6